A degree-three counterexample for the normalized upper Laplacian

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Abstract

Does a low codimension-one degree force zero into the spectrum of the normalized upper Laplacian of an infinite simplicial complex? We construct a countable pure three-dimensional complex with maximum triangle degree three whose upper Laplacian has spectrum bounded away from zero on the full degree-weighted triangle cochain space. This disproves both assertions in AIM Problem 643, including the stronger claim that degree at most the dimension forces a nonzero square-summable kernel. Every edge of the example has infinite triangle incidence. The proof uses disjoint square-summable fillings to control selected triangle values, then the gradient estimate on a binary tree to control all remaining coordinates.

1 The endpoint questions

The upper simplicial Laplacian measures the failure of a cochain to be a cocycle. Its normalized form also arises in the expected signed particle counts of simplicial branching random walks. Rosenthal computes the spectral measure on regular arboreal complexes and asks whether the corresponding endpoint eigenvalue persists for general low-degree complexes [2, Theorem 1.1 and Conjecture 4.3].

We use the precise formulation in AIM Problem 643 [1]. For a nonempty pure d-dimensional simplicial complex X, with $d \geq 2$, let $\deg(F)$ be the number of d-simplices containing a $(d-1)$-simplex F, and set $D = \sup_F \deg(F)$. The full cochain spaces are

$$H = \ell^2(X^{d-1}, \deg), \qquad K = \ell^2(X^d), \qquad \|f\|_H^2 = \sum_F \deg(F)|f(F)|^2,$$

using one orientation per simplex and counting measure in K. The usual coboundary is

$$(\delta f)([x_0, \dots, x_d]) = \sum_{i=0}^{d} (-1)^i f([x_0, \dots, \widehat{x}_i, \dots, x_d]), \qquad L = \delta^* \delta.$$

The two assertions are that $D \leq d+1$ forces $0 \in \operatorname{Spec}(L)$, and that $D \leq d$ forces $\ker L \neq \{0\}$. The complex may be infinite, and no finite-incidence condition is imposed on lower-dimensional faces.

Theorem 1. *There exists a countable nonempty pure three-dimensional simplicial complex X with one triangle of degree one and every other triangle of degree three. Every edge belongs to infinitely many triangles. For its full spaces $H = \ell^2(X^2, \deg)$ and $K = \ell^2(X^3)$, the usual coboundary satisfies*

$$\|f\|_H^2 \leq C \, \|\delta f\|_K^2 \quad (f \in H), \qquad C = 12 + \frac{3(1+2\sqrt{3})^2}{(\sqrt{2}-1)^2}.$$

Consequently

$$\operatorname{Spec}(\delta^*\delta) \subset [C^{-1}, 4], \qquad \ker(\delta^*\delta) = \{0\}.$$

Here $D = 3 \le d + 1 = 4$, while zero is absent from the spectrum. Also $D = 3 \le d = 3$, while the square-summable kernel is zero. Thus the same example disproves each universal assertion separately.

The construction repeatedly attaches copies of an apex triangle and a binary tree. A unit flux splits equally at each tree vertex, giving a square-summable filling of the apex triangle. The fresh vertices make these fillings orthogonal across copies. Their pairings with δf control the triangle values used as attachment targets. The body coboundary equations then express the remaining triangle values through tree gradients, which have a uniform lower bound. This last step gives control of the full cochain space rather than only the target coordinates.

2 Construction and face degrees

Let $W = \{0,1\}^{<\omega}$ be the finite binary words, including the empty word o. The rooted binary tree G has oriented edges $w \to ws$, where $s \in \{0,1\}$. Its root has degree two and every other vertex has degree three; every vertex has two children. Write $|w|$ for word length.

We first describe one part of the construction. Given an ordered triangle $A = (a,b,c)$ and a fresh copy of G, with vertices v_w, insert

$$[a,b,v_w,v_{ws}], \quad [b,c,v_w,v_{ws}], \quad [c,a,v_w,v_{ws}] \qquad (w \in W,\ s \in \{0,1\}), \tag{1}$$

and the single tetrahedron $[a,b,c,v_o]$. The first three families will be called body tetrahedra and the last one the bridge. Each triangle $[p,v_w,v_{ws}]$, with $p \in \{a,b,c\}$, is then used as the ordered apex triangle of a new copy of this construction. This is repeated at every generation.

Here is a precise vertex labeling for the recursion. Set

$$\mathcal{B} = \{0,1,2\} \times W \times \{0,1\}, \qquad \mathcal{I} = \mathcal{B}^{<\omega}.$$

For every $J \in \mathcal{I}$ and $w \in W$, take a distinct vertex $v_{J,w}$. Take three further vertices r_0, r_1, r_2, with different tags. Define ordered apex triples recursively by

$$A_\varnothing = (r_0,r_1,r_2), \qquad A_{J*(p,w,s)} = (a_{J,p}, v_{J,w}, v_{J,ws}), \quad A_J = (a_{J,0}, a_{J,1}, a_{J,2}). \tag{2}$$

The apex vertices of J come from smaller generations, so they are distinct from its fresh vertices $v_{J,w}$; the three apex vertices are distinct by induction. Let X be the downward closure of the tetrahedra in (1), using A_J and $v_{J,w}$ for every J, together with their bridges. No higher-dimensional simplices are added. Thus X is a countable, nonempty, pure three-dimensional abstract simplicial complex.

The triangles first appearing in copy J have two forms:

$$E(J,p,w,s) = [a_{J,p}, v_{J,w}, v_{J,ws}], \tag{3}$$

$$V(J,pq,w) = [a_{J,p}, a_{J,q}, v_{J,w}], \qquad pq \in \{01,12,20\}. \tag{4}$$

The bridge has the additional triangle $F_J = [A_J]$. For $J = \varnothing$ this is $F_0 = [r_0,r_1,r_2]$; otherwise it is the E triangle specified by the last letter of J. We use (3)–(4) as global triangle orientations. An inherited target F_J keeps its existing orientation, as prescribed by (2). The displayed tetrahedra fix their own orientations. Other orderings have the usual alternating sign.

Lemma 2. *Every triangle of X has a unique first appearance as F_0, E, or V. The tetrahedron sets of different copies are disjoint. Each E triangle is the target F_J of exactly one child copy. The triangle F_0 has degree one and all E and V triangles have degree three. Every edge of X belongs to infinitely many triangles.*

Proof. A new E or V triangle contains a vertex with the current copy label J. No smaller-generation copy contains that vertex. New faces in a different copy of the same generation have its different label, while their apex vertices come from smaller generations. New faces at later generations contain later labels. Within one copy, an E triangle has two new vertices and a V triangle has one; the apex vertex or edge and tree coordinates distinguish them. The only old triangle in a copy is its bridge target. The same argument distinguishes tetrahedra: bodies have two new vertices, bridges have one, and different tree edges and apex edges distinguish bodies. Formula (2) therefore gives a bijection between E triangles and noninitial copy targets. No V triangle is targeted.

An E triangle has two body cofaces in its birth copy and one bridge coface in its child. A V triangle at the tree root has two body cofaces and the bridge; away from the root it has three body cofaces. These are all cofaces, since a later tetrahedron can contain an old triangle only as its bridge target. The initial triangle has only the initial bridge coface.

Every apex edge has the distinct V cofaces indexed by all $w \in W$. A tree edge, or an edge joining an apex vertex to a tree vertex, lies in an E triangle and hence becomes an apex edge in the corresponding child. These exhaust the edge types in X, proving the last assertion. □

3 Square-summable fillings

Put $m(F) = \deg(F)$. Let H be the full space of complex triangle cochains with degree-weighted norm, and let K be the counting ℓ^2 space of tetrahedron cochains. Lemma 2 gives

$$\|f\|_H^2 = |f(F_0)|^2 + 3\sum_E |f(E)|^2 + 3\sum_V |f(V)|^2. \tag{5}$$

All sums use one chosen orientation per simplex. Inner products are linear in the first variable. On finitely supported cochains, Cauchy–Schwarz on the four facets of a tetrahedron gives

$$\|\delta f\|_K^2 \le 4\sum_F m(F)|f(F)|^2 = 4\,\|f\|_H^2\,. \tag{6}$$

Consequently δ extends to all H with norm at most two. Continuous coordinate evaluations show that its extension is still the usual alternating facet sum. Its bounded adjoint is

$$(\delta^* h)(F) = \frac{1}{m(F)} \sum_{T \supset F} \epsilon(T,F) h(T), \tag{7}$$

where $\epsilon(T,F)$ is the boundary incidence sign. Each sum has at most three terms. To verify membership in H without any lower-face assumption, pair the right side with finitely supported f. The finite rearrangement equals $\langle \delta f, h\rangle_K$ and is bounded by $2\,\|f\|_H\,\|h\|_K$. Taking f to be finite restrictions of that right side shows its H norm is at most $2\,\|h\|_K$; density then proves (7). In particular $L = \delta^*\delta$ is bounded, self-adjoint and positive, with $\|L\| \le 4$.

We next fill the target F_J in its own copy. Fix J temporarily, suppress its label, and write $A = [a,b,c]$ and $z = [a,b] + [b,c] + [c,a]$. For $N \ge 0$, let c_N be the finite tree one-chain having coefficient 2^{-n} on every edge whose child has depth $n \le N$. With counting norms on graph edges,

$$\partial c_N = -[v_o] + 2^{-N} \sum_{|w|=N} [v_w], \qquad \|c_N\|^2 = \sum_{n=1}^{N} 2^{-n} = 1 - 2^{-N}. \tag{8}$$

This includes $c_0 = 0$. At an interior vertex the incoming coefficient equals the sum of the two outgoing coefficients.

For finite chains, use $*$ to concatenate ordered vertices, extending bilinearly. Directly,

$$\partial[p,q,u,v] = [q,u,v] - [p,u,v] + [p,q,v] - [p,q,u].$$

Summing over the three apex cycle edges cancels the first two terms. Also $\partial[a,b,c,v_o] = z*[v_o] - A$. Therefore the finite top chain $h_{J,N} = -z * c_N - A * [v_o]$, identified with a cochain in K, satisfies

$$\partial h_{J,N} = F_J - 2^{-N} z * \sum_{|w|=N} [v_w], \qquad \|h_{J,N}\|_K^2 = 4 - 3\,2^{-N}. \tag{9}$$

Only finite formal boundaries have been used. The tail squared norm after depth N is $3\,2^{-N}$, so $h_{J,N}$ converges strongly in K to a cochain h_J with $\|h_J\|_K^2 = 4$.

The $3 \cdot 2^N$ leaf triangles in (9) are distinct V triangles of actual final degree three, also when $N = 0$. If $\mathbf{1}_F$ denotes the cochain supported at F with value one, (7) gives

$$\delta^* h_{J,N} = \frac{\mathbf{1}_{F_J}}{m(F_J)} + r_{J,N}, \qquad \|r_{J,N}\|_H^2 = (3 \cdot 2^N)\, 3 \left(\frac{2^{-N}}{3}\right)^2 = 2^{-N}.$$

The boundedness of δ^* now proves

$$\delta^* h_J = \frac{\mathbf{1}_{F_J}}{m(F_J)}, \qquad \|h_J\|_K^2 = 4. \tag{10}$$

Thus no formal boundary at an infinitely incident edge is needed.

4 A lower bound on the full cochain space

The cochains h_J have disjoint tetrahedron supports by Lemma 2. Hence $h_J/2$ form an orthonormal family in K. For every $f \in H$, (10) gives

$$\langle \delta f, h_J \rangle_K = \langle f, \mathbf{1}_{F_J}/m(F_J) \rangle_H = f(F_J).$$

Subtracting the orthogonal projection onto any finite subfamily from δf and expanding its squared norm gives the finite Bessel inequality. Increasing those finite sets to all J, and recalling that the targets are exactly F_0 and all E triangles, yields

$$|f(F_0)|^2 + \sum_E |f(E)|^2 \le 4\,\|\delta f\|_K^2\,, \tag{11}$$

$$|f(F_0)|^2 + 3\sum_E |f(E)|^2 \le 12\,\|\delta f\|_K^2\,. \tag{12}$$

It remains to control the V coordinates. Within each copy, pull tree coordinates back along $w \mapsto v_{J,w}$ and write

$$e_p(w, ws) = f([a_{J,p}, v_{J,w}, v_{J,ws}]), \qquad g_{pq}(w) = f([a_{J,p}, a_{J,q}, v_{J,w}]), \quad pq \in \{01, 12, 20\}.$$

Let e and g denote their counting ℓ^2 direct sums over all copies and indicated apex vertices or edges. The unique first appearances of the faces ensure that these are precisely the E and V coordinates, without duplication. In particular (11) gives $\|e\| \le 2\,\|\delta f\|_K$.

Let y be δf restricted to body tetrahedra, so $\|y\| \le \|\delta f\|_K$. Their alternating boundaries give

$$y_{pq}(w, ws) = g_{pq}(ws) - g_{pq}(w) + e_q(w, ws) - e_p(w, ws). \tag{13}$$

For each tree edge, the last three apex differences are the matrix

$$B = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix}$$

applied to (e_0, e_1, e_2). Multiplication gives $B^*B = 3I - \mathbf{J}_3$, where $\mathbf{J}_3$ is the all-ones matrix. Thus $\|Bv\|^2 = 3\|v\|^2 - |v_0 + v_1 + v_2|^2 \le 3\|v\|^2$. Taking the direct sum of these matrices in (13), we obtain

$$\|\nabla g\| \le \|y\| + \|Be\| \le (1 + 2\sqrt{3})\,\|\delta f\|_K\,, \qquad (\nabla g)_{pq}(w, ws) = g_{pq}(ws) - g_{pq}(w). \tag{14}$$

For a scalar $\ell^2(W)$ function q, index tree edges by their child and let Sq and Tq be the child and parent values respectively. Then

$$\|Sq\|^2 = \|q\|^2 - |q(o)|^2, \qquad \|Tq\|^2 = 2\,\|q\|^2\,.$$

The reverse triangle inequality implies $\|Sq - Tq\| \ge (\sqrt{2} - 1)\,\|q\|$. The same identities hold on the direct sum over all copies and apex edges, with the root terms summed. Hence

$$3\sum_V |f(V)|^2 = 3\,\|g\|^2 \le \frac{3(1 + 2\sqrt{3})^2}{(\sqrt{2} - 1)^2}\,\|\delta f\|_K^2\,. \tag{15}$$

Combining (5), (12) and (15) proves

$$\|f\|_H^2 \le C\,\|\delta f\|_K^2 \quad \text{for every } f \in H, \qquad C = 12 + \frac{3(1 + 2\sqrt{3})^2}{(\sqrt{2} - 1)^2}. \tag{16}$$

All direct-sum maps above are bounded and all estimates use the actual final face degrees. In particular the estimate applies to the whole H.

Completion of the proof of Theorem 1. Put $\alpha = C^{-1}$. Equations (6) and (16) give

$$\alpha\,\|f\|_H^2 \le \langle Lf, f\rangle_H \le 4\,\|f\|_H^2\,. \tag{17}$$

For completeness we verify the resolvent conclusion directly. A bounded Hilbert-space operator T satisfying positive lower bounds for both T and T^* is invertible: a convergent sequence Tx_n makes x_n Cauchy, so the range is closed, and $(\operatorname{ran} T)^\perp = \ker T^* = \{0\}$. A closed subspace with zero orthogonal complement is the whole space. Indeed, for a proper closed subspace M and $v \notin M$, let $\rho = \inf_{m \in M} \|v - m\| > 0$ and choose a minimizing sequence m_n, with $a_n = \|v - m_n\| \to \rho$. The parallelogram identity gives

$$\|m_n - m_\ell\|^2 \le 2(a_n^2 - \rho^2) + 2(a_\ell^2 - \rho^2).$$

It therefore converges to $m \in M$ with $\|v - m\| = \rho$. Varying m by real and imaginary multiples of a vector in M shows that $v - m$ is a nonzero vector in $M^\perp$. The lower bound for T also bounds its inverse.

For nonreal λ, the imaginary part of $\langle (L - \lambda I)f, f\rangle_H$ gives a lower bound $|\operatorname{Im}\lambda|\,\|f\|_H$ for $\|(L - \lambda I)f\|_H$, and likewise for its adjoint $L - \overline{\lambda} I$. For real $\lambda < \alpha$, (17) gives the lower bound $\alpha - \lambda$ for $L - \lambda I$ and its adjoint; for $\lambda > 4$ it gives the bound $\lambda - 4$ for $\lambda I - L$. The criterion just proved shows that all such λ are resolvent points. Thus $\operatorname{Spec}(L) \subset [\alpha, 4]$ and $\ker L = \{0\}$. Lemma 2 supplies all remaining geometric assertions. □

References

[1] MColbrook/AIM problem repository, *643. Zero in the upper-Laplacian spectrum of low-degree simplicial complexes*, statement at revision `a153f469`, `https://github.com/MColbrook/AIM/blob/a153f469ee0ea64e876b2ad97c79d252c19a3e22/problems/643-low-degree-simplicial-zero-spectrum.md`.

[2] R. Rosenthal, *Simplicial branching random walks and their applications*, arXiv:1412.5406v1 (2014), `https://arxiv.org/abs/1412.5406v1`.

AI use. The construction, proof development and manuscript preparation used OpenAI Codex. Natural-language checking was also AI-assisted. Independent mathematical review is requested.